

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5063B

5 Credits: 4

Program Elective

Source-grounded

Textile Machine Maintenance

□ To impart the knowledge of various Maintenance concepts , Erection procedures,

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5063B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5063B.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Textile Technology
Course code	5063B
Course title	Textile Machine Maintenance
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- To impart the knowledge of various Maintenance concepts , Erection procedures,
- application of lubricants, bearing and tools in textile industry.
- To develop the idea of maintenance schedule of different textile machines, its
- influence on product quality and the role of humidification plant, compressor, 5S
- system in manufacturing process.

Course outcomes

CO	Official outcome	Study level
CO1	operation, organisation structure and the role of 14 Understanding lubricants, bearing and tools in textile industry. Demonstrate gauges & equipments related to maintenance, erection procedure of textile	See official syllabus
CO2	15 Applying machines, vibration, maintenance of card clothing and top roller. Discuss the maintenance schedule of Preparatory and spinning machines, its influence	See official syllabus
CO3	on quality and the role of humidification plant, 14 Applying compressor and safety equipments in spinning Demonstrate the maintenance schedule of weaving preparatory, weaving and processing	See official syllabus
CO4	15 Applying machines, its influence on quality and the role of	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 1
CO2	CO2 3

Row	Extracted official mapping
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	structure and the role of lubricants, bearing and tools in textile	4	As specified
Module 2	of textile machines, vibration, maintenance of card clothing and top	4	As specified
Module 3	machines, its influence on quality and the role of humidification plant,	4	As specified
Module 4	and processing machines, its influence on quality and the role of 5S.	4	As specified

MODULE 1

structure and the role of lubricants, bearing and tools in textile

This module is presented from the official syllabus scope for Textile Machine Maintenance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: structure and the role of lubricants, bearing and tools in textile
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding maintenance Understand the organization structure of

3 Understanding maintenance Understand the organization structure of is an official topic in Module 1 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding maintenance Understand the organization structure of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding maintenance understand the organization structure of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding maintenance Understand the organization structure of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding maintenance
Understand the organization structure of മെന്റേനൻസ് ഓഫീസ്
definition/meaning മെന്റേനൻസ്. മെന്റേനൻസ് തരങ്ങൾ, steps,
application മെന്റേനൻസ് പ്രവർത്തനം. Technical terms English-ൽ മെന്റേനൻസ്
മെന്റേനൻസ്.

▼ maintenance department and duties of 3 Understanding maintenance personnel Understand the function and types of

maintenance department and duties of 3 Understanding maintenance personnel Understand the function and types of is an official topic in Module 1 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify maintenance department and duties of 3 Understanding maintenance personnel Understand the function and types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, maintenance department and duties of 3 understanding maintenance personnel understand the function and types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What maintenance department and duties of 3 Understanding maintenance personnel Understand the function and types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റൂട്ടിംഗ് മെയിന്റനൻസ് ഡിപ്പാർട്ട്മെന്റ് ആൻഡ് ഡ്യൂട്ടീസ് ഓഫ് 3 Understanding maintenance personnel Understand the function and types of റൂട്ടിംഗ് മെയിന്റനൻസ് ഡിപ്പാർട്ട്മെന്റ്. റൂട്ടിംഗ് മെയിന്റനൻസ് ഡിപ്പാർട്ട്മെന്റ്, steps, application റൂട്ടിംഗ് മെയിന്റനൻസ് ഡിപ്പാർട്ട്മെന്റ്. Technical terms English-റൂട്ടിംഗ് മെയിന്റനൻസ് ഡിപ്പാർട്ട്മെന്റ്.

▼ 4 Understanding lubricants. Understand the types of bearing and its

4 Understanding lubricants. Understand the types of bearing and its is an official topic in Module 1 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding lubricants. Understand the types of bearing and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding lubricants. understand the types of bearing and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding lubricants. Understand the types of bearing and its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 4 Understanding lubricants. Understand the types of bearing and its definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 4 Understanding application in textile machines Define maintenance - objectives - 2 systems of maintenance - Draw back & benefits - Types of maintenance - Routine, Preventive, Predictive maintenance, Remedial, Restorative maintenance, Emergency maintenance- Examples of different maintenance activities in spinning mill under each type - Maintenance man power norms per 1000 spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms - General Duties of Maintenance supervisor - Define lubricants - functions - broad Classification - Properties of liquid lubricant - Define ISO VG classification of industrial oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box - Need for grease and solid lubricants - Commonly used solid lubricants - Different Oil and grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing - Structure - function - Classification of bearings based on Rolling element -Types of Ball bearings and Roller bearings - Causes of bearing failure - Application of bearing in textile machines - Maintenance of chain, gear and gear box. Illustrate the application of gauges & equipments, erection procedure

4 Understanding application in textile machines Define maintenance - objectives - 2 systems of maintenance - Draw back & benefits - Types of maintenance - Routine, Preventive, Predictive maintenance, Remedial, Restorative maintenance, Emergency maintenance- Examples of different maintenance activities in spinning mill under each type - Maintenance man power norms per 1000 spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms - General Duties of Maintenance supervisor - Define lubricants - functions - broad Classification - Properties of liquid lubricant - Define ISO VG classification of industrial oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box - Need for grease and solid lubricants - Commonly used solid lubricants - Different Oil and grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing - Structure - function - Classification of bearings based on Rolling

element -Types of Ball bearings and Roller bearings - Causes of bearing failure - Application of bearing in textile machines - Maintenance of chain, gear and gear box. Illustrate the application of gauges & equipments, erection procedure is an official topic in Module 1 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding application in textile machines Define maintenance - objectives - 2 systems of maintenance - Draw back & benefits - Types of maintenance - Routine, Preventive, Predictive maintenance, Remedial, Restorative maintenance, Emergency maintenance- Examples of different maintenance activities in spinning mill under each type - Maintenance man power norms per 1000 spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms - General Duties of Maintenance supervisor - Define lubricants - functions - broad Classification - Properties of liquid lubricant - Define ISO VG classification of industrial oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box - Need for grease and solid lubricants - Commonly used solid lubricants - Different Oil and grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing - Structure - function - Classification of bearings based on Rolling element -Types of Ball bearings and Roller bearings - Causes of bearing failure - Application of bearing in textile machines - Maintenance of chain, gear and gear box. Illustrate the application of gauges & equipments, erection procedure using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding application in textile machines define maintenance - objectives - 2 systems of maintenance - draw back & benefits - types of maintenance - routine, preventive, predictive maintenance, remedial, restorative maintenance, emergency maintenance- examples of different maintenance activities in spinning mill under each type - maintenance man power norms per 1000 spindles for cotton mill - maintenance organization structure of 50,000 spindleage mill and weaving department with 100 looms - general duties of maintenance supervisor - define lubricants - functions - broad classification - properties of liquid lubricant - define iso vg classification of industrial oil based on viscosity grade - factors influencing selection of lubricating oil for gear box - need for grease and solid lubricants - commonly used solid lubricants - different oil and grease grade, viscosity in cst, their area of application in spinning mill - define bearing - structure - function - classification of bearings based on rolling element -types of ball bearings and roller bearings - causes of bearing failure - application of bearing in textile machines - maintenance of chain, gear and gear box. illustrate the application of gauges & equipments, erection procedure should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding application in textile machines Define maintenance - objectives - 2 systems of maintenance - Draw back & benefits - Types of maintenance - Routine, Preventive, Predictive maintenance, Remedial, Restorative maintenance, Emergency maintenance- Examples of different maintenance activities in spinning mill under each type - Maintenance man power norms per 1000 spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms - General Duties of Maintenance supervisor - Define lubricants - functions - broad

Aspect	What to prepare
	Classification - Properties of liquid lubricant - Define ISO VG classification of industrial oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box - Need for grease and solid lubricants - Commonly used solid lubricants - Different Oil and grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing - Structure - function - Classification of bearings based on Rolling element -Types of Ball bearings and Roller bearings - Causes of bearing failure - Application of bearing in textile machines - Maintenance of chain, gear and gear box. Illustrate the application of gauges & equipments, erection procedure means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding application in textile machines Define maintenance - objectives - 2 systems of maintenance - Draw back & benefits - Types of maintenance - Routine, Preventive, Predictive maintenance, Remedial, Restorative maintenance, Emergency maintenance- Examples of different maintenance activities in spinning mill under each type - Maintenance man power norms per 1000 spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms - General Duties of Maintenance supervisor - Define lubricants - functions - broad Classification - Properties of liquid lubricant - Define ISO VG classification of industrial oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box - Need for grease and solid lubricants - Commonly used solid lubricants -

Different Oil and grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing - Structure - function - Classification of bearings based on Rolling element -Types of Ball bearings and Roller bearings - Causes of bearing failure - Application of bearing in textile machines - Maintenance of chain, gear and gear box. Illustrate the application of gauges & equipments, erection procedure
 definition/meaning . steps, application . Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

of textile machines, vibration, maintenance of card clothing and top

This module is presented from the official syllabus scope for Textile Machine Maintenance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of textile machines, vibration, maintenance of card clothing and top
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ vibration, damping ,balancing and function 4 Remembering of the levelling instruments Understand the role of fasteners, general

vibration, damping ,balancing and function 4 Remembering of the levelling instruments Understand the role of fasteners, general is an official topic in Module 2 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify vibration, damping ,balancing and function 4 Remembering of the levelling instruments Understand the role of fasteners, general using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, vibration, damping ,balancing and function 4 remembering of the levelling instruments understand the role of fasteners, general should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What vibration, damping ,balancing and function 4 Remembering of the levelling instruments Understand the role of fasteners, general means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-വൈബ്രേഷൻ, ഡാമ്പിംഗ്, ബാലൻസിംഗ് and function 4 Remembering of the levelling instruments Understand the role of fasteners, general ഏകദേശം definition/meaning. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ൽ ഏകദേശം ഏകദേശം.

▼ tools, equipments for maintenance and 4 Understanding erection

Discuss the erection procedure of textile

tools, equipments for maintenance and 4 Understanding erection Discuss the erection procedure of textile is an official topic in Module 2 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify tools, equipments for maintenance and 4 Understanding erection Discuss the erection procedure of textile using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, tools, equipments for maintenance and 4 understanding erection discuss the erection procedure of textile should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What tools, equipments for maintenance and 4 Understanding erection Discuss the erection procedure of textile means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്ന tools, equipments for maintenance and 4 Understanding erection Discuss the erection procedure of textile ഫാബ്രിക്കേഷൻ എന്നതിന്റെ definition/meaning എഴുതുക. ഇതിന്റെ പ്രധാന ഘട്ടങ്ങൾ, steps, application വിശദീകരിക്കുക. Technical terms English-ൽ പറയുക.

▼ 4 Understanding machines Illustrate the maintenance of card clothing

4 Understanding machines Illustrate the maintenance of card clothing is an official topic in Module 2 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding machines Illustrate the maintenance of card clothing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding machines illustrate the maintenance of card clothing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding machines Illustrate the maintenance of card clothing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 4 Understanding machines Illustrate the maintenance of card clothing. definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 3 Understanding and top roller Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of vibration - Causes of vibrations - Damping and its methods - Define balancing and its types - Define Temporary fasteners - Role of each type - function of oil seal and gaskets - General Tools for maintenance and erection - Types of spanner and its available sizes, Torque wrench, Allen key, Hammers and its types, File, Circlip Pliers, Oil cans, Grease gun - General gauges for maintenance - Construction and working of Roller eccentricity tester - Function of ATIRA Nilometer, TARP gauge, Tachometer, Feeler gauge, Depth gauge, Plug gauge, Vernier calliper, Micrometer. General erection equipments - General points of erection - Arranging components & Work table - Provisions for power and pneumatic lines - Floor leveling - Machine layout line marking - Basic Erection sequence - Positioning the base machine - leveling & settings - Commissioning - Trial run - Basic erection procedure - carding machine, Ring frame, Power loom - Card Wire mounting, Stripping and Grinding machines -Top Roller Maintenance - Cots mounting - buffing - Berkolisation - Testing of Top Roller Concentricity, Shore Hardness and Surface Roughness. Discuss the maintenance schedule of Preparatory and spinning

3 Understanding and top roller Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of vibration - Causes of vibrations - Damping and its methods - Define balancing and its types - Define Temporary fasteners - Role of each type - function of oil seal and gaskets - General Tools for maintenance and erection - Types of spanner and its available sizes, Torque wrench, Allen key, Hammers and its types, File, Circlip Pliers, Oil cans, Grease gun - General gauges for maintenance - Construction and working of Roller eccentricity tester - Function of ATIRA Nilometer, TARP gauge, Tachometer, Feeler gauge, Depth gauge, Plug gauge, Vernier calliper, Micrometer. General erection equipments - General points of erection - Arranging components & Work table - Provisions for

power and pneumatic lines - Floor leveling - Machine layout line marking - Basic Erection sequence - Positioning the base machine - leveling & settings - Commissioning - Trial run - Basic erection procedure - carding machine, Ring frame, Power loom - Card Wire mounting, Stripping and Grinding machines -Top Roller Maintenance - Cots mounting - buffing - Berkolisation - Testing of Top Roller Concentricity, Shore Hardness and Surface Roughness. Discuss the maintenance schedule of Preparatory and spinning is an official topic in Module 2 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and top roller Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of vibration - Causes of vibrations - Damping and its methods - Define balancing and its types - Define Temporary fasteners - Role of each type - function of oil seal and gaskets - General Tools for maintenance and erection - Types of spanner and its available sizes, Torque wrench, Allen key, Hammers and its types, File, Circlip Pliers, Oil cans, Grease gun - General gauges for maintenance - Construction and working of Roller eccentricity tester - Function of ATIRA Nilometer, TARP gauge, Tachometer, Feeler gauge, Depth gauge, Plug gauge, Vernier calliper, Micrometer. General erection equipments - General points of erection - Arranging components & Work table - Provisions for power and pneumatic lines - Floor leveling - Machine layout line marking - Basic Erection sequence - Positioning the base machine - leveling & settings - Commissioning - Trial run - Basic erection procedure - carding machine, Ring frame, Power loom - Card Wire mounting, Stripping and Grinding

machines -Top Roller Maintenance - Cots mounting - buffing - Berkolisation - Testing of Top Roller Concentricity, Shore Hardness and Surface Roughness. Discuss the maintenance schedule of Preparatory and spinning using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and top roller purpose of leveling - role of foundation - construction and working of spirit level - role of plumb bob, wedge with spirit level, tri square - vibration - types of vibration - causes of vibrations - damping and its methods - define balancing and its types - define temporary fasteners - role of each type - function of oil seal and gaskets - general tools for maintenance and erection - types of spanner and its available sizes, torque wrench, allen key, hammers and its types, file, circlip pliers, oil cans, grease gun - general gauges for maintenance - construction and working of roller eccentricity tester - function of atira nilometer, tarp gauge, tachometer, feeler gauge, depth gauge, plug gauge, vernier calliper, micrometer. general erection equipments - general points of erection - arranging components & work table - provisions for power and pneumatic lines - floor leveling - machine layout line marking - basic erection sequence - positioning the base machine - leveling & settings - commissioning - trial run - basic erection procedure - carding machine, ring frame, power loom - card wire mounting, stripping and grinding machines -top roller maintenance - cots mounting - buffing - berkolisation - testing of top roller concentricity, shore hardness and surface roughness. discuss the maintenance schedule of preparatory and spinning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding and top roller Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of vibration - Causes of vibrations - Damping and its methods - Define balancing and its types - Define Temporary fasteners - Role of each type - function of oil seal and gaskets - General Tools for maintenance and erection - Types of spanner and its available sizes, Torque wrench, Allen key, Hammers and its types, File, Circlip Pliers, Oil cans, Grease gun - General gauges for maintenance - Construction and working of Roller eccentricity tester - Function of ATIRA Nilometer, TARP gauge, Tachometer, Feeler gauge, Depth gauge, Plug gauge, Vernier calliper, Micrometer. General erection equipments - General points of erection - Arranging components & Work table - Provisions for power and pneumatic lines - Floor leveling - Machine layout line marking - Basic Erection sequence - Positioning the base machine - leveling & settings - Commissioning - Trial run - Basic erection procedure - carding machine, Ring frame, Power loom - Card Wire mounting, Stripping and Grinding machines -Top Roller Maintenance - Cots mounting - buffing - Berkolisation - Testing of Top Roller Concentricity, Shore Hardness and Surface Roughness. Discuss the maintenance schedule of Preparatory and spinning means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Understanding and top roller Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of

vibration - Causes of vibrations - Damping and its methods - Define balancing and its types - Define Temporary fasteners - Role of each type - function of oil seal and gaskets - General Tools for maintenance and erection - Types of spanner and its available sizes, Torque wrench, Allen key, Hammers and its types, File, Circlip Pliers, Oil cans, Grease gun - General gauges for maintenance - Construction and working of Roller eccentricity tester - Function of ATIRA Nilometer, TARP gauge, Tachometer, Feeler gauge, Depth gauge, Plug gauge, Vernier calliper, Micrometer. General erection equipments - General points of erection - Arranging components & Work table - Provisions for power and pneumatic lines - Floor leveling - Machine layout line marking - Basic Erection sequence - Positioning the base machine - leveling & settings - Commissioning - Trial run - Basic erection procedure - carding machine, Ring frame, Power loom - Card Wire mounting, Stripping and Grinding machines -Top Roller Maintenance - Cots mounting - buffing - Berkolisation - Testing of Top Roller Concentricity, Shore Hardness and Surface Roughness. Discuss the maintenance schedule of Preparatory and spinning
 definition/meaning .
 , steps, application
 . Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

machines, its influence on quality and the role of humidification plant,

This module is presented from the official syllabus scope for Textile Machine Maintenance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: machines, its influence on quality and the role of humidification plant,
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Remembering of machines used in spinning mill. Illustrate the defects due to improper

3 Remembering of machines used in spinning mill. Illustrate the defects due to improper is an official topic in Module 3 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering of machines used in spinning mill. Illustrate the defects due to improper using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering of machines used in spinning mill. illustrate the defects due to improper should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Remembering of machines used in spinning mill. Illustrate the defects due to improper means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Remembering of machines used in spinning mill. Illustrate the defects due to improper definition/meaning. , steps, application . Technical terms English- .

▼ 4 Understanding maintenance of spinning machines Discuss the role of humidification plant and

4 Understanding maintenance of spinning machines Discuss the role of humidification plant and is an official topic in Module 3 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding maintenance of spinning machines Discuss the role of humidification plant and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding maintenance of spinning machines discuss the role of humidification plant and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding maintenance of spinning machines Discuss the role of humidification plant and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 4 Understanding maintenance of spinning machines Discuss the role of humidification plant and definition/meaning . steps, application . Technical terms English- .

▼ 4 Understanding compressor in spinning mill Summarize the causes of accidents in textile

4 Understanding compressor in spinning mill Summarize the causes of accidents in textile is an official topic in Module 3 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding compressor in spinning mill Summarize the causes of accidents in textile using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding compressor in spinning mill summarize the causes of accidents in textile should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding compressor in spinning mill Summarize the causes of accidents in textile means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 4 Understanding compressor in spinning mill Summarize the causes of accidents in textile definition/meaning, steps, application Technical terms English-

▼ 3 Understanding mills and the role of safety equipments Routine maintenance schedule of Blow room, Carding, Drawing, Combing, Speed frame, Ring frame and Auto coner - Defects due to improper maintenance - Low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank CV %, Ratching in speed frame, More Yarn end breakage in Ring spinning - Need for controlling temperature and humidity - Recommended R H in different departments of cotton mill- Lay out of humidification plant - Function of each parts- Damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - Flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure , compressor capacity) - Air receiver, dryer, Pressure drop and its control. Causes of accidents in textile mills - Protective equipment for head, eye, face, ear, hands, foot and respiratory - general safety - fire alarms, extinguishers Illustrate the maintenance schedule of weaving preparatory, weaving

3 Understanding mills and the role of safety equipments Routine maintenance schedule of Blow room, Carding, Drawing, Combing, Speed frame, Ring frame and Auto coner - Defects due to improper maintenance - Low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank CV %, Ratching in speed frame, More Yarn end breakage in Ring spinning - Need for controlling temperature and humidity - Recommended R H in different departments of cotton mill- Lay out of humidification plant - Function of each parts- Damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - Flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure , compressor capacity) - Air receiver, dryer, Pressure drop and its control. Causes of accidents in textile mills - Protective equipment for head, eye, face, ear, hands, foot and respiratory -

general safety - fire alarms, extinguishers Illustrate the maintenance schedule of weaving preparatory, weaving is an official topic in Module 3 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding mills and the role of safety equipments
Routine maintenance schedule of Blow room, Carding, Drawing, Combing, Speed frame, Ring frame and Auto coner - Defects due to improper maintenance - Low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank CV %, Ratching in speed frame, More Yarn end breakage in Ring spinning - Need for controlling temperature and humidity - Recommended R H in different departments of cotton mill- Lay out of humidification plant - Function of each parts- Damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - Flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure , compressor capacity) - Air receiver, dryer, Pressure drop and its control. Causes of accidents in textile mills - Protective equipment for head, eye, face, ear, hands, foot and respiratory - general safety - fire alarms, extinguishers Illustrate the maintenance schedule of weaving preparatory, weaving using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mills and the role of safety equipments routine maintenance schedule of blow room, carding, drawing, combing, speed frame, ring frame and auto coner - defects due to improper maintenance - low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank cv %, ratching in speed frame, more yarn end breakage in ring spinning - need for controlling temperature and humidity - recommended r h in different departments of cotton mill- lay out of humidification plant - function of each parts- damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure , compressor capacity) - air receiver, dryer, pressure drop and its control. causes of accidents in textile mills - protective equipment for head, eye, face, ear, hands, foot and respiratory - general safety - fire alarms, extinguishers illustrate the maintenance schedule of weaving preparatory, weaving should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding mills and the role of safety equipments Routine maintenance schedule of Blow room, Carding, Drawing, Combing, Speed frame, Ring frame and Auto coner - Defects due to improper maintenance - Low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank CV %, Ratching in speed frame, More Yarn end breakage in Ring spinning - Need for controlling temperature and humidity - Recommended R H in different departments of cotton mill- Lay out of humidification plant - Function of each parts- Damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - Flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure ,

Aspect	What to prepare
	compressor capacity) - Air receiver, dryer, Pressure drop and its control. Causes of accidents in textile mills - Protective equipment for head, eye, face, ear, hands, foot and respiratory - general safety - fire alarms, extinguishers Illustrate the maintenance schedule of weaving preparatory, weaving means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding mills and the role of safety equipments Routine maintenance schedule of Blow room, Carding, Drawing, Combing, Speed frame, Ring frame and Auto coner - Defects due to improper maintenance - Low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank CV %, Ratching in speed frame, More Yarn end breakage in Ring spinning - Need for controlling temperature and humidity - Recommended R H in different departments of cotton mill- Lay out of humidification plant - Function of each parts- Damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - Flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure , compressor capacity) - Air receiver, dryer, Pressure drop and its control. Causes of accidents in textile mills - Protective equipment for head, eye, face, ear, hands, foot and respiratory - general safety - fire alarms, extinguishers Illustrate the maintenance schedule of weaving preparatory, weaving definition/meaning

പ്രായോഗികമായി. പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ, steps, application പദങ്ങൾ
പ്രായോഗികമായി പദങ്ങൾ. Technical terms English-പദങ്ങൾ പദങ്ങൾ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

and processing machines, its influence on quality and the role of 5S.

This module is presented from the official syllabus scope for Textile Machine Maintenance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and processing machines, its influence on quality and the role of 5S.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ of weaving preparatory and weaving 3 Remembering machine. Illustrate the defects due to improper

of weaving preparatory and weaving 3 Remembering machine. Illustrate the defects due to improper is an official topic in Module 4 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify of weaving preparatory and weaving 3 Remembering machine. Illustrate the defects due to improper using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, of weaving preparatory and weaving 3 remembering machine. illustrate the defects due to improper should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What of weaving preparatory and weaving 3 Remembering machine. Illustrate the defects due to improper means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- of weaving preparatory and weaving 3 Remembering machine. Illustrate the defects due to improper definition/meaning . steps, application . Technical terms English- .

▼ 4 Understanding maintenance. Understand the maintenance schedule of

4 Understanding maintenance. Understand the maintenance schedule of is an official topic in Module 4 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding maintenance. Understand the maintenance schedule of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding maintenance. understand the maintenance schedule of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding maintenance. Understand the maintenance schedule of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്ന 4 Understanding maintenance. Understand the maintenance schedule of ഏതെങ്കിലും ഒരു വാഹനം definition/meaning എന്തെന്നും. എങ്ങനെ പരിപാലിക്കണം, steps, application എങ്ങനെ പരിപാലിക്കണം എന്നും. Technical terms English-ൽ എഴുതണം.

▼ 4 Understanding processing machines Understand the concept of Housekeeping,

4 Understanding processing machines Understand the concept of Housekeeping, is an official topic in Module 4 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding processing machines Understand the concept of Housekeeping, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding processing machines understand the concept of housekeeping, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding processing machines Understand the concept of Housekeeping, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Understanding processing machines Understand the concept of Housekeeping, definition/meaning. , steps, application . Technical terms English- .

▼ 4 Applying 5S and its advantages in textile industry Maintenance schedule of Pirn winding machine, Direct Warping , Sizing machine, High speed automatic shuttle Loom, Rapier and projectile loom - Care and maintenance of loom parts Heald, Reed, Picker, Shuttle, Buffer - Defects due to improper maintenance of preparatory machines and looms . Sizing machine - more ends out of lease , excessive size droppings, Lapping on squeeze rolls and warp beams, sticky beam , Shuttle loom- machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile Up keep of Boilers and Kiers, Padding mangles, Stenter - Maintenance schedule of Dyeing machine - Introduction to 5S work practices - steps involved in implementation of 5S - Advantages of 5S in textile industry - Points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams. Text / Reference: T/R Book Title/Author Neeraj Niiijaawan and Rashmi Niiijaawan - Modern approach to T1 maintenance in Spinning Ratnam T V, Ramachandran M and Narayanaswamy G -Maintenance R2 management in spinning - SITRA R3 SITRA - Maintenance management in Weaving Joshi B B, Vora K B and Chetvani H G, "Spinning, Weaving, Processing R4 machinery maintenance in Textile Mills", R5 Textile machinery manufacturers Erection manuals and handouts Online resources: Sl.No Website Link 1 www.textilemates.com 2 <https://textilelearner.blogspot.com>

4 Applying 5S and its advantages in textile industry Maintenance schedule of Pirn winding machine, Direct Warping , Sizing machine, High speed automatic shuttle Loom, Rapier and projectile loom - Care and maintenance of loom parts Heald, Reed, Picker, Shuttle, Buffer - Defects due to improper maintenance of preparatory machines and looms . Sizing machine - more ends out of lease , excessive size droppings, Lapping on squeeze rolls and warp beams, sticky beam , Shuttle loom- machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile Up keep of Boilers and Kiers, Padding mangles, Stenter - Maintenance schedule of Dyeing machine - Introduction to 5S work practices - steps involved in implementation of 5S - Advantages

of 5S in textile industry - Points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams. Text / Reference: T/R Book Title/Author Neeraj Nijjaawan and Rashmi Nijjaawan - Modern approach to T1 maintenance in Spinning Ratnam T V, Ramachandran M and Narayanaswamy G -Maintenance R2 management in spinning - SITRA R3 SITRA - Maintenance management in Weaving Joshi B B, Vora K B and Chetvani H G, "Spinning, Weaving, Processing R4 machinery maintenance in Textile Mills", R5 Textile machinery manufacturers Erection manuals and handouts Online resources: Sl.No Website Link 1 www.textilemates.com 2 <https://textilelearner.blogspot.com> is an official topic in Module 4 of Textile Machine Maintenance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying 5S and its advantages in textile industry Maintenance schedule of Pirn winding machine, Direct Warping , Sizing machine, High speed automatic shuttle Loom, Rapier and projectile loom - Care and maintenance of loom parts Heald, Reed, Picker, Shuttle, Buffer - Defects due to improper maintenance of preparatory machines and looms . Sizing machine - more ends out of lease , excessive size droppings, Lapping on squeeze rolls and warp beams, sticky beam , Shuttle loom- machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile Up keep of Boilers and Kiers, Padding mangles, Stenter - Maintenance schedule of Dyeing machine - Introduction to 5S work practices - steps involved in implementation of 5S - Advantages of 5S in textile industry - Points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams. Text / Reference: T/R

Book Title/Author Neeraj Nijjaawan and Rashmi Nijjaawan - Modern approach to T1 maintenance in Spinning Ratnam T V, Ramachandran M and Narayanaswamy G -Maintenance R2 management in spinning – SITRA R3 SITRA - Maintenance management in Weaving Joshi B B, Vora K B and Chetvani H G, "Spinning, Weaving, Processing R4 machinery maintenance in Textile Mills", R5 Textile machinery manufacturers Erection manuals and handouts Online resources: Sl.No Website Link 1 www.textilemates.com 2 <https://textilelearner.blogspot.com> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying 5s and its advantages in textile industry maintenance schedule of pirn winding machine, direct warping , sizing machine, high speed automatic shuttle loom, rapier and projectile loom - care and maintenance of loom parts heald, reed, picker, shuttle, buffer - defects due to improper maintenance of preparatory machines and looms . sizing machine - more ends out of lease , excessive size droppings, lapping on squeeze rolls and warp beams, sticky beam , shuttle loom- machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile up keep of boilers and kiers, padding mangles, stenter - maintenance schedule of dyeing machine - introduction to 5s work practices - steps involved in implementation of 5s - advantages of 5s in textile industry - points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams. text / reference: t/r book title/author neeraj nijjaawan and rashmi nijjaawan - modern approach to t1 maintenance in spinning ratnam t v, ramachandran m and narayanaswamy g -maintenance r2 management in spinning – sitra r3 sitra - maintenance management in weaving joshi b b, vora k b and chetvani h g, "spinning, weaving, processing r4 machinery maintenance in textile mills", r5 textile machinery manufacturers erection manuals and handouts online resources: sl.no website link 1 www.textilemates.com 2 <https://textilelearner.blogspot.com> should be discussed with attention to safe

operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying 5S and its advantages in textile industry Maintenance schedule of Pirn winding machine, Direct Warping , Sizing machine, High speed automatic shuttle Loom, Rapier and projectile loom - Care and maintenance of loom parts Heald, Reed, Picker, Shuttle, Buffer - Defects due to improper maintenance of preparatory machines and looms . Sizing machine - more ends out of lease , excessive size droppings, Lapping on squeeze rolls and warp beams, sticky beam , Shuttle loom-machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile Up keep of Boilers and Kiers, Padding mangles, Stenter - Maintenance schedule of Dyeing machine - Introduction to 5S work practices - steps involved in implementation of 5S - Advantages of 5S in textile industry - Points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams. Text / Reference: T/R Book Title/Author Neeraj Nijjaawan and Rashmi Nijjaawan - Modern approach to T1 maintenance in Spinning Ratnam T V, Ramachandran M and Narayanaswamy G - Maintenance R2 management in spinning - SITRA R3 SITRA - Maintenance management in Weaving Joshi B B, Vora K B and Chetvani H G, "Spinning, Weaving, Processing R4 machinery maintenance in Textile Mills", R5 Textile machinery manufacturers Erection manuals and handouts Online resources: Sl.No Website Link 1 www.textilemates.com 2 https://textilelearner.blogspot.com means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Applying 5S and its advantages in textile industry Maintenance schedule of Pirn winding machine, Direct Warping , Sizing machine, High speed automatic shuttle Loom, Rapier and projectile loom - Care and maintenance of loom parts Heald, Reed, Picker, Shuttle, Buffer - Defects due to improper maintenance of preparatory machines and looms . Sizing machine - more ends out of lease , excessive size droppings, Lapping on squeeze rolls and warp beams, sticky beam , Shuttle loom- machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile Up keep of Boilers and Kiers, Padding mangles, Stenter - Maintenance schedule of Dyeing machine - Introduction to 5S work practices - steps involved in implementation of 5S - Advantages of 5S in textile industry - Points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams. Text / Reference: T/R Book Title/Author Neeraj Nijjaawan and Rashmi Nijjaawan - Modern approach to T1 maintenance in Spinning Ratnam T V, Ramachandran M and Narayanaswamy G -Maintenance R2 management in spinning – SITRA R3 SITRA - Maintenance management in Weaving Joshi B B, Vora K B and Chetvani H G, "Spinning, Weaving, Processing R4 machinery maintenance in Textile Mills", R5 Textile machinery manufacturers Erection manuals and handouts Online resources: Sl.No Website Link 1 www.textilemates.com 2 <https://textilelearner.blogspot.com> definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

- in spinning mill under each type - Maintenance man power norms per 1000
- spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms -
- General Duties of Maintenance supervisor - Define lubricants - functions - broad
- Classification - Properties of liquid lubricant - Define ISO VG classification of industrial
- oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box -
- Need for grease and solid lubricants - Commonly used solid lubricants - Different Oil and
- grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing -
- Structure - function - Classification of bearings based on Rolling element -Types of Ball
- bearings and Roller bearings - Causes of bearing failure - Application of bearing in textile
- machines - Maintenance of chain, gear and gear box.
- Illustrate the application of gauges & equipments, erection procedure
- CO2 of textile machines, vibration, maintenance of card clothing and top Understand the concept of leveling,
- M2.01 vibration, damping ,balancing and function 4 Remembering of the levelling instruments Understand the role of fasteners, general
- M2.02 tools, equipments for maintenance and 4 Understanding erection Discuss the erection procedure of textile
- M2.03 4 Understanding machines Illustrate the maintenance of card clothing
- M2.04 3 Understanding and top roller
- Series Test I 1
- Contents:
- Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role

- of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of vibration - Causes

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 3 Understanding maintenance Understand the organization structure of. (3 marks)**

► **Define or explain maintenance department and duties of 3 Understanding maintenance personnel Understand the function and types of. (3 marks)**

► **Define or explain 4 Understanding lubricants. Understand the types of bearing and its. (3 marks)**

► **Define or explain 4 Understanding application in textile machines**
Define maintenance - objectives - 2 systems of maintenance - Draw back & benefits - Types of maintenance - Routine, Preventive, Predictive maintenance, Remedial, Restorative maintenance, Emergency maintenance- Examples of different maintenance activities in spinning mill under each type - Maintenance man power norms per 1000 spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms - General Duties of Maintenance supervisor - Define lubricants - functions - broad Classification - Properties of liquid lubricant - Define ISO VG classification of industrial oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box - Need for grease and solid lubricants - Commonly used solid lubricants - Different Oil and grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing - Structure - function - Classification of bearings based on Rolling element -Types of Ball bearings and Roller bearings - Causes

of bearing failure - Application of bearing in textile machines - Maintenance of chain, gear and gear box. Illustrate the application of gauges & equipments, erection procedure. (3 marks)

Module 2

► Define or explain vibration, damping ,balancing and function 4 Remembering of the levelling instruments Understand the role of fasteners, general. (3 marks)

► Define or explain tools, equipments for maintenance and 4 Understanding erection Discuss the erection procedure of textile. (3 marks)

► Define or explain 4 Understanding machines Illustrate the maintenance of card clothing. (3 marks)

► Define or explain 3 Understanding and top roller Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of vibration - Causes of vibrations - Damping and its methods - Define balancing and its types - Define Temporary fasteners - Role of each type - function of oil seal and gaskets - General Tools for maintenance and erection - Types of spanner and its available sizes, Torque wrench, Allen key, Hammers and its types, File, Circlip Pliers, Oil cans, Grease gun - General gauges for maintenance - Construction and working of Roller eccentricity tester - Function of ATIRA Nilometer, TARP gauge, Tachometer, Feeler gauge, Depth gauge, Plug gauge, Vernier calliper, Micrometer. General erection equipments - General points of erection - Arranging components & Work table - Provisions for power and pneumatic lines - Floor leveling - Machine layout line marking - Basic Erection sequence - Positioning the base machine - leveling & settings - Commissioning - Trial run - Basic erection procedure - carding machine, Ring frame, Power loom - Card Wire mounting, Stripping and Grinding machines -Top Roller Maintenance - Cots mounting - buffing -

Berkolisation - Testing of Top Roller Concentricity, Shore Hardness and Surface Roughness. Discuss the maintenance schedule of Preparatory and spinning. (3 marks)

Module 3

► **Define or explain 3 Remembering of machines used in spinning mill. Illustrate the defects due to improper. (3 marks)**

► **Define or explain 4 Understanding maintenance of spinning machines Discuss the role of humidification plant and. (3 marks)**

► **Define or explain 4 Understanding compressor in spinning mill Summarize the causes of accidents in textile. (3 marks)**

► **Define or explain 3 Understanding mills and the role of safety equipments Routine maintenance schedule of Blow room, Carding, Drawing, Combing, Speed frame, Ring frame and Auto coner - Defects due to improper maintenance - Low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank CV %, Ratching in speed frame, More Yarn end breakage in Ring spinning - Need for controlling temperature and humidity - Recommended R H in different departments of cotton mill- Lay out of humidification plant - Function of each parts- Damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - Flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure , compressor capacity) - Air receiver, dryer, Pressure drop and its control. Causes of accidents in textile mills - Protective equipment for head, eye, face, ear, hands, foot and respiratory - general safety - fire alarms, extinguishers Illustrate the maintenance schedule of weaving preparatory, weaving. (3 marks)**

Module 4

- ▶ **Define or explain of weaving preparatory and weaving 3 Remembering machine. Illustrate the defects due to improper. (3 marks)**
- ▶ **Define or explain 4 Understanding maintenance. Understand the maintenance schedule of. (3 marks)**
- ▶ **Define or explain 4 Understanding processing machines Understand the concept of Housekeeping,. (3 marks)**
- ▶ **Define or explain 4 Applying 5S and its advantages in textile industry Maintenance schedule of Pirn winding machine, Direct Warping , Sizing machine, High speed automatic shuttle Loom, Rapier and projectile loom - Care and maintenance of loom parts Heald, Reed, Picker, Shuttle, Buffer - Defects due to improper maintenance of preparatory machines and looms . Sizing machine - more ends out of lease , excessive size droppings, Lapping on squeeze rolls and warp beams, sticky beam , Shuttle loom- machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile Up keep of Boilers and Kiers, Padding mangles, Stenter - Maintenance schedule of Dyeing machine - Introduction to 5S work practices - steps involved in implementation of 5S - Advantages of 5S in textile industry - Points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams. Text / Reference: T/R Book Title/Author Neeraj Nijjaawan and Rashmi Nijjaawan - Modern approach to T1 maintenance in Spinning Ratnam T V, Ramachandran M and Narayanaswamy G - Maintenance R2 management in spinning - SITRA R3 SITRA - Maintenance management in Weaving Joshi B B, Vora K B and Chetvani H G, "Spinning, Weaving, Processing R4 machinery maintenance in Textile Mills", R5 Textile machinery manufacturers Erection manuals and handouts Online resources: Sl.No Website Link 1 www.textilemates.com 2 <https://textilelearner.blogspot.com>. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding maintenance Understand the organization structure of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **vibration, damping ,balancing and function 4 Remembering of the levelling instruments Understand the role of fasteners, general.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Remembering of machines used in spinning mill. Illustrate the defects due to improper.**

Module 4

1-mark definition: State the meaning of **of weaving preparatory and weaving 3 Remembering**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

machine. Illustrate the defects due to improper.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link www.textilemates.com <https://textilelearner.blogspot.com>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5063B. Verify institutional updates against the current official curriculum.